Nawel AMARA

Computer Engineering Student | Full-Stack & AI-Integrated Systems

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Profile

Computer Engineering student with strong hands-on experience in full-stack development, backend systems, and data-driven applications. Passionate about Artificial Intelligence and intelligent systems, with practical experience integrating AI features into real-world applications. Skilled in building scalable, efficient, and user-centered solutions. Seeking international internship and research opportunities to apply and expand expertise in AI, software engineering, and data systems.

Experience

TEEP Research Intern - AI New Media Lab, 09/2026 – Present
National Taichung University of Eductation (NTCU) Taiwan Taichung

- Support research in Artificial Intelligence, Deep Learning, Computer Vision, and Multimodal AI.
- Assist with data preprocessing, dataset preparation, model development, and experimental evaluation.
- Explore Large Language Models (LLMs), BERT-based text analysis, and Vision-Language Models. - Conduct literature reviews, analyze research findings, and contribute to technical documentation.
- Contribute to AI-driven digital content and intelligent visual systems within an interdisciplinary research laboratory.

Final-Year Project — Intelligent Master Data Governance Platform, 02/2026 – 05/2026
Leoni tunisia sousse

Tech stack: Spring Boot, Angular, PostgreSQL , LangChain4j , JWT, RESTful APIs

- Engineered a full-stack platform digitizing Excel-based industrial data validation workflows across multiple data entity types, eliminating manual entry errors and improving traceability.
- Designed and developed a RAG-powered AI assistant using RAG architecture and LangChain4j, enabling users to query project data and workflow records through natural language instead of manual spreadsheet searches.
- Designed a JWT-secured Audit Trail module providing end-to-end traceability of all data modifications meeting industrial compliance requirements.
- Developed flexible PostgreSQL schemas to handle heterogeneous industrial data structures while maintaining referential consistency.

Summer Internship — Experience & Reservation Module(E-Tourism Platform), *DotCom, Nabeul* 06/2025 – 09/2025
Nabeul

Tech stack: Node.js, Angular, PostgreSQL (JSONB), Git, SSH

- Designed and integrated a full Experience & Reservation module into a production international e-tourism platform, supporting dynamic availability management, participant caps, and booking conflict resolution.

- Modeled PostgreSQL JSONB schemas with business constraints to ensure data consistency across multilingual experience listings.
- Implemented backend logic for availability checks, real-time reservation flows, and participant management — deployed via SSH to a remote VM.
- Contributed to Angular frontend features including search, filters, and booking flows in an Agile team environment.

MERN Quiz Platform (Academic Project), *EPI Digital School, sousse* 03/2025 – 05/2025
Tech stack: MongoDB, Express.js, React, Node.js, JWT sousse

- Built a full-stack quiz platform with role-based access control (admin / student), JWT authentication, and voucher-gated quiz access.
- Implemented real-time quiz participation and scoring workflows with RESTful API integration and responsive React UI.

Final-Year Project - ERP Timesheet Module, *B2M-IT* 02/2023 – 06/2023
Tech stack: Microsoft Dynamics 365 ,Power BI ,ML integration Nabeul

- Developed a custom timesheet and project management module for employee time tracking within a Dynamics 365 ERP environment.
- Integrated BI dashboards and ML-based decision-support models for project management reporting.

Internship — E-Commerce Platform, *Hammamet Valley Hub, Hammamet* 07/2022 – 08/2022
Tech stack: Symfony ,MySQL Hammamet

- Built a complete e-commerce web application including product catalog, cart, and order management.

Education

National Engineering Degree in Computer Engineering, 09/2023 – 06/2026
EPI-international multidisciplinary school Sousse

Applied Bachelor's Degree in Computer Technologies, 2020 – 2023
Higher Institute of Technological Studies of Kelibia Nabeul

Baccalaureate Diploma in Mathematics, 09/2019 – 06/2020
Mohamed Boudhina High School, Hammamet Nabeul Hammamet Nabeul

Technical Proficiencies

Software Development
Angular, React.js, Node.js, Express.js, RESTful APIs, Spring Boot ,Microservices basics

AI & Data
RAG systems, LangChain4j, LLM integration, Scikit-learn, TensorFlow/Keras, Pandas, NumPy

Programming Languages
Java, JavaScript, TypeScript, Python

Databases
PostgreSQL, MongoDB, MySQL, Firebase Firestore

DevOps & Tools
Git, Docker, GitHub Actions (basics), Postman, SSH/VM deployment

Languages

Arabic
Native

English
TOEIC Certificate
Professional

French
DELF B2 Certificate
Professional

Certificates

- AWS Academy Graduate—AWS Academy Cloud Foundation
- CCNAv7: Introduction to Networks
- CCNAv7: Switching, Routing, and Wireless Essentials